

Lesson 1.1 – Introduction to Pruning

What is Pruning?

Pruning is the practice of removing unnecessary or unwanted branches, shoots, or leaves to improve plant health and productivity.

Key Goal

Improve structure

Increase sunlight penetration

Enhance fruit quality

Example

A mango tree with too many branches:

Inner branches block sunlight → fruits stay small

After pruning → sunlight reaches all parts → better fruit size and sweetness

Module 2: Pruning Tools & Equipment

Lesson 2.1 – Choosing the Right Tool

Recommended Tools:

Bypass pruners → for clean cuts on live branches

Pruning saw → for thicker branches

Why Not Dull Tools?

Cause ragged cuts

Increase risk of disease infection

Demo (Step-by-step)

Select a sharp bypass pruner

Position it at a 45° angle

Cut smoothly in one motion

Module 3: Understanding Tree Structure

Lesson 3.1 – Branch Problems

Crossed or Rubbing Branches

These branches:

Damage each other

Create wounds → entry points for disease

Example

Two branches rubbing:

Bark gets stripped

Fungus enters → branch dies

What to Do

Remove the weaker or poorly positioned branch

Module 4: Pruning Techniques

Lesson 4.1 – Thinning Cuts

Definition:

Removing a branch completely at its origin.

Purpose:

Improves airflow

Allows sunlight inside canopy

Demo

Locate overcrowded area

Identify entire branch to remove

Cut at branch collar (not flush)

Lesson 4.2 – Heading Cuts

Definition:

Shortening a branch without removing it entirely.

Effect:

Stimulates new growth near the cut

Example

Cutting the tip of a branch:

New shoots grow below → bushier plant

Module 5: Proper Cutting Techniques

Lesson 5.1 – Cutting at the Branch Collar

Correct Method:

Cut just outside the branch collar

Do NOT:

Cut flush with trunk

Leave long stubs

Why?

Trees heal naturally through compartmentalization

Demo

✓ Correct: slight outward cut

✗ Wrong: flat cut against trunk

Module 6: Tree Health Management

Lesson 6.1 – Removing Dead or Diseased Wood

Always remove:

Dead branches

Diseased parts

Damaged limbs

Example

A branch with fungus:

If left → spreads to entire tree

If removed early → tree survives

Lesson 6.2 – Water Sprouts & Suckers

Water Sprouts

Fast-growing upright shoots

Waste energy

Suckers (below graft union)

Grow from rootstock

Do NOT produce desired fruit

Action

Remove at the base immediately

Module 7: Pruning Timing

Lesson 7.1 – Seasonal Pruning

Winter (Dormant Season)

Best for structure shaping

Summer Pruning

Controls excessive growth

Maintains shape

Lesson 7.2 – Flowering Plants

Plants that bloom on old wood:

→ Prune after flowering

Example

If pruned before blooming:

You remove flower buds → no flowers that season

Module 8: Growth Control & Fertilization

Lesson 8.1 – Effects of Over-Fertilizing

Too much nitrogen causes:

Excessive weak growth

More pruning needed

Higher pest risk

Module 9: Safe Pruning Practices

Lesson 9.1 – Safety Guidelines

Always:

Wear eye protection

Use stable ladder

Be aware of falling branches

Never:

Prune on wet surfaces

Use dull tools

Work carelessly overhead

Module 10: Advanced Pruning Strategies

Lesson 10.1 – Canopy Management Rule

Golden Rule:

Remove only 25%–33% per year

Why?

Prevents stress

Maintains tree health

Lesson 10.2 – Restorative Pruning

For overgrown trees:

Prune gradually over several seasons

Example

Bad approach:

Cutting 80% → tree shock or death

Correct approach:

Year 1: remove 30%

Year 2: reshape

Year 3: maintain

Module 11: Tool Hygiene

Lesson 11.1 – Sanitizing Tools

Important when:

Disease is present

Demo

Dip tool in alcohol or bleach solution

Wipe clean

Continue pruning

Module 12: Myths & Misconceptions

Lesson 12.1 – Wound Sealants

Common Myth:

Trees need paint or sealant

Reality:

Trees heal naturally

Sealants can trap moisture → cause decay